

EE6211-POWER ELECTRONICS LAB

Experiment 6 Report

submitted by

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Contents

1	Experiment 6	1
1.1	Aim	1
1.2	Software	1
1.3	Theory	1
1.3.1	DC-DC Boost Converter	1
1.3.2	Actual Voltage Conversion Ratio	2
1.3.3	Component Design and Selection	2
1.3.4	State-Space Averaging and Small-Signal Modeling	3
1.3.5	Loop Shaping Control Design	6
1.3.6	Gate Driver Circuit	6
1.3.7	Sensing Circuits	6
1.3.8	Snubber Circuit	7
1.3.9	Protection Elements	7
1.4	Circuit Diagram	8
2	Calculations	10
2.1	Specifications Considered	10
2.2	Converter Calculations	10
2.2.1	Input Power and Current	10
2.2.2	Duty Cycle	11
2.2.3	Inductor Design	12
2.2.4	Capacitor Design	12
2.2.5	MOSFET Ratings	14
2.2.6	Snubber Design	15
2.2.7	Miscellaneous Calculations	16
2.3	Sensing Circuit Calculations	17
2.3.1	Voltage Sensor	17
2.3.2	Current Sensor	18
2.3.3	R-C Filter	18
2.3.4	Op-Amp Calculations	19
2.3.5	ADC Input Filter	19
2.4	PCB Considerations	20

2.5	Controller Calculations	20
2.5.1	DC Operating Point	20
2.5.2	Control-to-Output Transfer Function	21
2.5.3	Loop Shaping Design	22
3	Results	24
4	Learnings/Observations	25

1. Experiment 6

1.1 Aim

The objective is to design and analyze a synchronous DC–DC boost converter operating over an input range of 48V – 62V, delivering 64V – 85V at up to 200W, with a switching frequency of 80kHz and an efficiency of 95%. The design includes component selection (inductor, capacitors, MOSFETs) while limiting inductor current ripple to 20% and output voltage ripple to 2%. A small-signal model is derived using state-space averaging, to obtain the output voltage control transfer function. A PI controller is designed via loop-shaping and validated in MATLAB/Simulink, comparing open- and closed-loop performance across voltage references.

Building on the theoretical design, a practical implementation is realized using a C2000 microcontroller for closed-loop digital control. Hardware development covers PCB layout, magnetic design, capacitor and semiconductor selection, sensing and conditioning circuits, isolated gate drivers, and snubber design.

1.2 Software

1. MATLAB/Simulink
2. Simscape Specialized Power Systems Library
3. KiCAD EDA

1.3 Theory

1.3.1 DC-DC Boost Converter

A boost converter is a switch-mode power supply topology that produces an output voltage greater than its input voltage. It operates on the principle of energy storage and transfer through an inductor, controlled by a electronic switch, switched at high frequencies. The converter consists of an inductor L , a lower side MOSFET, an upper

side MOSFET, and an output capacitor C connected to the load resistance R. The switching action is governed by a periodic gate signal at frequency f_{sw} , dividing each switching period into two sub-intervals defined by the duty cycle D.

During the ON interval of duration DT_s , the MOSFET conducts and the inductor stores energy from the input source. During the OFF interval of duration $(1 - D)T_s$, the lower side MOSFET turns off, the upper side MOSFET conducts, and the stored inductor energy is transferred to the output capacitor and load.

The fundamental volt-second balance applied to the inductor and charge balance applied to the capacitor yield the ideal steady-state voltage conversion ratio:

$$\frac{V_{out}}{V_{in}} = \frac{1}{1 - D} \quad (1.1)$$

1.3.2 Actual Voltage Conversion Ratio

The issue with the ideal case ratio is that the gain approaches infinity as the duty cycle goes to one. To account for this aberration, the inductor parasitic resistance is considered. Applying the Volt-Sec balance with r_l included gives:

$$(V_{in} - r_l I_L) \cdot D = V_{out} \cdot (1 - D) \quad (1.2)$$

Substituting for I_L from the Current-Sec balance condition,

$$(I_L - I_{out}) \cdot (1 - D) = I_{out} \cdot D \quad (1.3)$$

$$I_L = \frac{I_{out}}{1 - D} \quad (1.4)$$

The final expression is given by Eq. ??.

$$\frac{V_{out}}{V_{in}} = \frac{R}{r_L + R \cdot (1 - D)^2} \times (1 - D) \quad (1.5)$$

This shows that the actual output voltage is always lower than the ideal value, with the deviation growing at higher duty cycles and heavier loads. The corrected duty cycle must therefore be determined by solving the resulting quadratic equation to achieve the desired output voltage, given below:

$$1 - D = \frac{V_{in}}{2V_{out}} + \sqrt{\left(\frac{V_{in}}{2V_{out}}\right)^2 - \frac{r_L}{R}} \quad (1.6)$$

1.3.3 Component Design and Selection

The inductor value is determined from the volt-second balance during the ON interval, ensuring that the peak-to-peak inductor current ripple Δi_L does not exceed 20% of the nominal input current at any operating point:

$$L = \frac{V_{in} D}{\Delta i_L f_{sw}} \quad (1.7)$$

The output capacitor is sized to limit the output voltage ripple ΔV_{out} to 2% of the output voltage:

$$C = \frac{I_{out}D}{\Delta V_{out}f_{sw}} \quad (1.8)$$

Power device ratings are determined from the worst case operating conditions. The MOSFET is rated based on its RMS current during the ON interval:

$$I_{SW,rms} = I_L\sqrt{D} \quad (1.9)$$

The freewheeling diode is rated based on its average current during the OFF interval:

$$I_{D,avg} = (1 - D)I_L \quad (1.10)$$

Both devices must block a voltage equal to V_{out} when not conducting. A voltage derating factor of 30% is applied to both devices.

1.3.4 State-Space Averaging and Small-Signal Modeling

The State-Space Averaging (SSA) technique derives a continuous-time dynamic model of the switched converter by averaging the state equations over one switching period. The boost converter state variables are the inductor current i_L and capacitor voltage v_C , with the state vector $\mathbf{x} = [i_L, v_C]^T$, input $\mathbf{u} = [v_{in}]^T$, and output $y = v_{out}$. The general state-space representation is given by:

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u} \quad (1.11)$$

$$\mathbf{y} = \mathbf{C}\mathbf{x} + \mathbf{D}\mathbf{u} \quad (1.12)$$

Interval 1 — Switch ON (Duration DT)

During the ON interval, the MOSFET conducts and the diode is reverse biased. Applying KVL around the inductor loop and KCL at the output node gives:

$$\frac{di_L}{dt} = \frac{V_{in}}{L} - \frac{r_L}{L}i_L \quad (1.13)$$

$$\frac{dv_C}{dt} = -\frac{v_C}{RC} \quad (1.14)$$

The state matrices for this interval are therefore:

$$A_1 = \begin{bmatrix} -\frac{r_L}{L} & 0 \\ 0 & -\frac{1}{RC} \end{bmatrix}, \quad B_1 = \begin{bmatrix} \frac{1}{L} \\ 0 \end{bmatrix}, \quad C_1 = [0 \quad 1], \quad D_1 = [0] \quad (1.15)$$

Interval 2 — Switch OFF (Duration $(1 - D)T$)

During the OFF interval, the MOSFET is open and the diode conducts. Applying KVL and KCL gives:

$$\frac{di_L}{dt} = \frac{V_{in}}{L} - \frac{r_L}{L}i_L - \frac{v_C}{L} \quad (1.16)$$

$$\frac{dv_C}{dt} = \frac{i_L}{C} - \frac{v_C}{RC} \quad (1.17)$$

The state matrices for this interval are therefore:

$$A_2 = \begin{bmatrix} -\frac{r_L}{L} & -\frac{1}{L} \\ \frac{1}{C} & -\frac{1}{RC} \end{bmatrix}, \quad B_2 = \begin{bmatrix} \frac{1}{L} \\ 0 \end{bmatrix}, \quad C_2 = [0 \quad 1], \quad D_2 = [0] \quad (1.18)$$

Averaged Model

The averaged state matrices are obtained by weighting each subinterval matrix by its respective duty cycle:

$$A = DA_1 + (1 - D)A_2, \quad B = DB_1 + (1 - D)B_2 \quad (1.19)$$

$$C = DC_1 + (1 - D)C_2, \quad D_{sys} = DD_1 + (1 - D)D_2 \quad (1.20)$$

Substituting and simplifying yields the averaged state matrices:

$$A = \begin{bmatrix} -\frac{r_L}{L} & -\frac{(1 - D)}{L} \\ \frac{(1 - D)}{C} & -\frac{1}{RC} \end{bmatrix}, \quad B = \begin{bmatrix} \frac{1}{L} \\ 0 \end{bmatrix}, \quad C = [0 \quad 1], \quad D_{sys} = [0] \quad (1.21)$$

The DC operating point is obtained by setting $\dot{\mathbf{x}} = 0$, giving $A\mathbf{X} + B\mathbf{U} = 0$. Solving yields:

$$I_L = \frac{V_{in}}{r_L + (1 - D)^2 R}, \quad V_C = \frac{(1 - D)R \cdot V_{in}}{r_L + (1 - D)^2 R} \quad (1.22)$$

Small-Signal Linearization

Linearizing around the DC operating point, all variables are perturbed as:

$$i_L = I_L + \hat{i}_L, \quad v_C = V_C + \hat{v}_C, \quad d = D + \hat{d}, \quad v_{in} = V_{in} + \hat{v}_{in} \quad (1.23)$$

Substituting into the averaged state equations, the DC terms cancel and nonlinear products of small-signal quantities ($\hat{x}\hat{y}$) are neglected, yielding:

$$\dot{\hat{x}} = [DA_1 + (1-D)A_2]\hat{x} + [DB_1 + (1-D)B_2]\hat{v}_{in} + [(A_1 - A_2)X + (B_1 - B_2)V_{in}]\hat{d} \quad (1.24)$$

This gives the small-signal state-space model:

$$\dot{\hat{x}} = A\hat{x} + B\hat{v}_{in} + E\hat{d} \quad (1.25)$$

$$\hat{y} = C\hat{x} + D\hat{u} \quad (1.26)$$

The small-signal input matrix E associated with the duty cycle perturbation $\hat{d}$ is computed as:

$$(A_1 - A_2) = \begin{bmatrix} 0 & \frac{1}{L} \\ -\frac{1}{C} & 0 \end{bmatrix}, \quad (B_1 - B_2) = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad (1.27)$$

$$E = (A_1 - A_2)X + (B_1 - B_2)V_{in} = \begin{bmatrix} 0 & \frac{1}{L} \\ -\frac{1}{C} & 0 \end{bmatrix} \begin{bmatrix} I_L \\ V_C \end{bmatrix} = \begin{bmatrix} \frac{V_C}{L} \\ -\frac{I_L}{C} \end{bmatrix} \quad (1.28)$$

The complete small-signal state-space representation is therefore:

$$\begin{bmatrix} \dot{\hat{i}}_L \\ \dot{\hat{v}}_C \end{bmatrix} = \begin{bmatrix} -\frac{r_L}{L} & -\frac{(1-D)}{L} \\ \frac{(1-D)}{C} & -\frac{1}{RC} \end{bmatrix} \begin{bmatrix} \hat{i}_L \\ \hat{v}_C \end{bmatrix} + \begin{bmatrix} \frac{1}{L} \\ 0 \end{bmatrix} \hat{v}_{in} + \begin{bmatrix} \frac{V_C}{L} \\ -\frac{I_L}{C} \end{bmatrix} \hat{d} \quad (1.29)$$

$$\hat{v}_{out} = \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} \hat{i}_L \\ \hat{v}_C \end{bmatrix} \quad (1.30)$$

Control-to-Output Transfer Function

The transfer functions relating each input to the output are obtained from the small-signal model as:

$$G_{vd}(s) = C(sI - A)^{-1}E, \quad G_{vg}(s) = C(sI - A)^{-1}B \quad (1.31)$$

Evaluating $G_{vd}(s)$ gives the control-to-output transfer function:

$$G_{vd}(s) = \frac{1}{LC} \frac{(1-D)V_C - I_L r_L - LI_L s}{s^2 + s \left(\frac{r_L}{L} + \frac{1}{RC} \right) + \frac{r_L + (1-D)^2 R}{RLC}} \quad (1.32)$$

1.3.5 Loop Shaping Control Design

Loop shaping is a frequency-domain controller design methodology in which the open-loop transfer function $L(s) = G_{vd}(s) \cdot C(s)$ is shaped in the Bode plot to meet desired stability and performance specifications. The design targets a specified phase margin (PM) and gain crossover frequency ω_c .

A PI controller of the form:

$$C(s) = K_p + \frac{K_i}{s} \quad (1.33)$$

is designed by evaluating the plant at the desired crossover frequency ω_c , computing the required compensator gain $K_C = 1/|G_{vd}(j\omega_c)|$ and phase angle:

$$\delta_c = PM - 180^\circ - \angle G_{vd}(j\omega_c) \quad (1.34)$$

The controller parameters are then extracted as:

$$K_p = K_c \cos(\delta_c), \quad K_i = -K_c \cdot \omega_c \cdot \sin(\delta_c) \quad (1.35)$$

1.3.6 Gate Driver Circuit

The gate driver circuit is responsible for driving the power MOSFETs by supplying the required gate voltage and current. Since, the MOSFET has internal capacitances, the rapid charging and discharging of these is necessary to achieve fast switching and reduce switching losses. An isolated gate driver is employed to provide galvanic isolation between the control and power stages, to protect the low-voltage circuitry from high-voltage transients.

The driver provides a gate voltage of 15V to ensure fast turn ON. The circuit is designed as per the specifications of the AMC1300 IC datasheet. The driver is placed in close proximity to the MOSFETs to minimize the loop inductance ensuring stable and efficient performance.

1.3.7 Sensing Circuits

Accurate sensing of the input, output voltages and the inductor current is essential for the reliable for the closed loop control of the converter. This report considers only the single loop control of the output voltage, but provisions for extending to a double or a triple loop control has been given by arranging an input voltage and current sensor circuits.

The output voltage is measured using a resistor divider network, which scales the high voltage to a level suitable with the micro-controller's ADC. Current sensing is implemented using low value shunt resistors, where the voltage developed across the shunt is proportional to the current. To ensure safety and noise rejection, an isolated amplifier is used in the sensing path, providing galvanic isolation between the power

stage and control circuitry.

The signal is conditioned using a differential amplifier to improve accuracy and reject common-mode noise. RC filtering is used to remove high-frequency switching noise.

1.3.8 Snubber Circuit

During switching transitions, parasitic inductances and capacitances in the circuit lead to voltage spikes and high-frequency oscillations at the switching node. These transients can stress the MOSFETs, reducing its life. To mitigate these effects, an RC snubber circuit is connected across the devices to damp the oscillations and limit voltage overshoot.

The snubber capacitor absorbs the transient energy, while the resistor dissipates it, thereby reducing ringing. Proper placement of the snubber close to the MOSFETs minimizes these parasitic effects. The snubber parameters are selected based on empirical expressions.

1.3.9 Protection Elements

Fuse

A fuse is incorporated at the input and output stage of the converter to enable over-current protection in the circuit. Under such abnormal conditions, excessive current can damage the components and pose safety risks.

The fuse rating is selected based on the maximum operating current with an appropriate safety margin. Placement at the terminals ensures that faults are isolated from the source and load, thereby protecting the converter, source and the load.

Varistor (MOV)

A metal oxide varistor is used to protect the converter from transient overvoltage conditions such as voltage surges and switching induced spikes. The MOV is characterized by a non-linear voltage-current relationship, remaining non-conductive when the voltage exceeds a specified threshold. The MOV clamps the voltage to a safe level by diverting the excess energy away from the sensitive components.

It is placed across input and output terminals to provide immediate suppression of transients. Proper selection of the clamping voltage and energy rating ensures effective protection without interfering with normal converter operation.

Bleeder Resistor

A bleeder resistor is connected across the terminals of the input and output electrolytic capacitors to ensure proper discharge when the system is powered down. This prevents high voltage from remaining for extended periods. It also attenuates the initial transients when the capacitors charge from zero.

Minimum Load Resistor

In DC-DC converters, operation under very light loads or no-load conditions can lead to the saturation of the inductor core and over-voltages at the output capacitor. To ensure stable operation, a minimum load is often required so that the converter remains within the designed operating region, typically continuous conduction mode.

1.4 Circuit Diagram

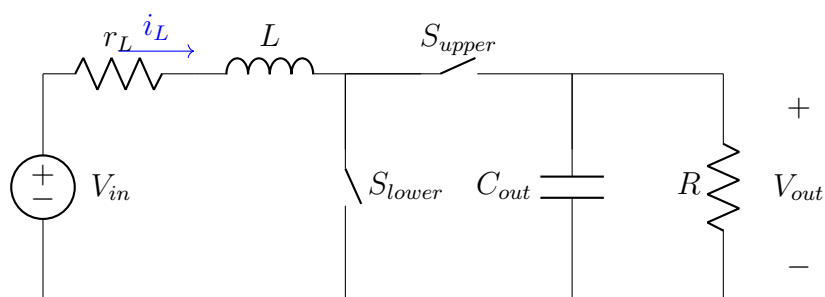


Figure 1.1: Boost Converter Main Circuit (basic)

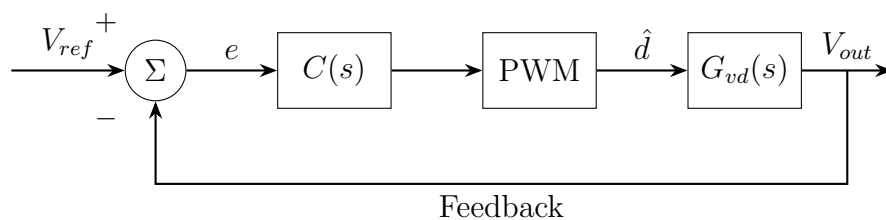


Figure 1.2: Closed-Loop Control Block Diagram

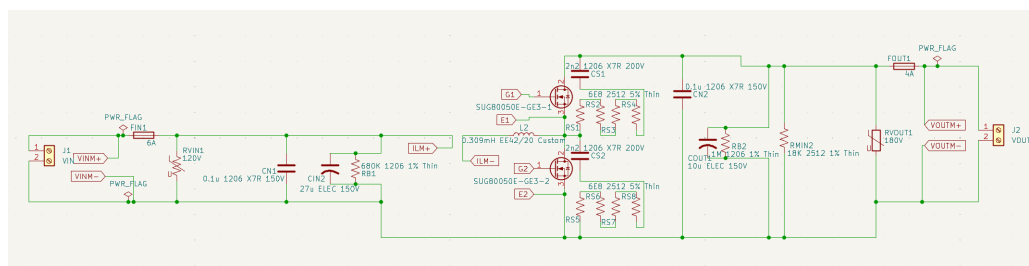
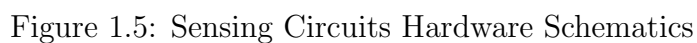


Figure 1.3: Boost Converter Circuit Hardware Schematic



2. Calculations

2.1 Specifications Considered

The boost converter is designed according to the specifications given in Table 2.1.

Parameter	Symbol	Value
Input Voltage	V_{in}	$48\text{ V} - 62\text{ V}$
Output Voltage	V_{out}	$64\text{ V} - 85\text{ V}$
Maximum Output Power	$P_{out_{max}}$	200 W
Switching Frequency	f_{sw}	80 kHz
Efficiency	η	95%
Current Ripple	$\frac{\Delta I_L}{I_L}$	20%
Voltage Ripple	$\frac{\Delta V_{out}}{V_{out}}$	2%
Inductor Parasitic Resistance	r_L	$10\text{m}\Omega$

Table 2.1: DC-DC Boost Converter Specifications

2.2 Converter Calculations

2.2.1 Input Power and Current

The maximum input power is determined from the output power and efficiency as:

$$P_{in} = \frac{P_{out_{max}}}{\eta} = \frac{200}{0.95} = 210.53\text{ W} \quad (2.1)$$

The input current, which is equal to the average inductor current, is given by:

$$I_L = I_{in} = \frac{P_{in}}{V_{in}} \quad (2.2)$$

The variation of input current with input voltage is given in Table 2.2. Since the inductor current is maximum at minimum input voltage, $V_{in} = 48 \text{ V}$ is used as the worst case design condition, giving $I_{in} = I_L = 4.386 \text{ A}$.

V_{in}	I_{in}
48 V	4.386 A
62 V	3.396 A

Table 2.2: Input Current Variation

2.2.2 Duty Cycle

The ideal duty cycle is derived from the voltage conversion ratio and is given by:

$$D_{ideal} = 1 - \frac{V_{in}}{V_{out}} \quad (2.3)$$

However, as discussed in Section 1.5, the actual duty cycle must account for the inductor parasitic resistance r_L , and is obtained by solving the quadratic expression:

$$1 - D_{actual} = \frac{V_{in}}{2V_{out}} + \sqrt{\left(\frac{V_{in}}{2V_{out}}\right)^2 - \frac{r_L}{R}} \quad (2.4)$$

The duty cycle is evaluated at all boundary combinations of input and output voltage for both the ideal and actual cases, and the results are listed in Table 2.3.

V_{in}	V_{out}	D_{ideal}	D_{actual}
48 V	64 V	0.250	0.251
	85 V	0.435	0.436
62 V	64 V	0.031	0.032
	85 V	0.271	0.271

Table 2.3: Ideal vs Actual Duty Cycle at Boundary Operating Points

2.2.3 Inductor Design

The peak-to-peak inductor current ripple must not exceed 20% of the nominal input current at any operating point. From the volt-second balance, the minimum inductance required is:

$$\frac{\Delta I_L}{I_L} \leq \frac{DV_{in}}{Lf_{sw}I_{in}} \quad (2.5)$$

Rearranging for the inductance:

$$L \geq \frac{DV_{in}}{f_{sw}I_{in} \frac{\Delta I_L}{I_L}} \quad (2.6)$$

The inductance is evaluated at each boundary duty cycle and the results are listed in Table 2.4. The worst case value of $L = 0.309 \text{ mH}$ corresponding to $V_{in} = 62 \text{ V}$, $V_{out} = 85 \text{ V}$ is selected for the design.

V_{in}	V_{out}	Duty Cycle	Inductance
62 V	64 V	0.031	0.035 mH
48 V	64 V	0.250	0.171 mH
48 V	85 V	0.435	0.298 mH
62 V	85 V	0.271	0.309 mH

Table 2.4: Inductance Values at Boundary Operating Points

The key current quantities used in subsequent design calculations are summarised in Table 2.5. A custom footprint was developed to include in the PCB considering the core dimensions.

2.2.4 Capacitor Design

Input Capacitor

The input capacitor must filter the discontinuous input current drawn by the converter and limit the peak-to-peak input voltage ripple to 2% of the input voltage. The approximated capacitance is given by:

$$C_{in} \geq \frac{I_{in} \cdot D}{\Delta V_{in} \cdot f_{sw}} \quad (2.7)$$

where $\Delta V_{in} = 0.02 \times V_{in}$. The capacitance is evaluated at all boundary conditions and the results are listed in Table 2.6.

Parameter	Formula	Value
I_L	$\frac{P_{in}}{V_{in_{min}}}$	4.386 A
ΔI_L	$\frac{\Delta I_L}{I_L} \cdot I_{in}$	0.877 A
I_L^{rms}	$\sqrt{I_L^2 + \frac{\Delta I_L^2}{12}}$	4.429 A
i_p	$I_L + \frac{\Delta I_L}{2}$	4.824 A

Table 2.5: Inductor Current Calculations

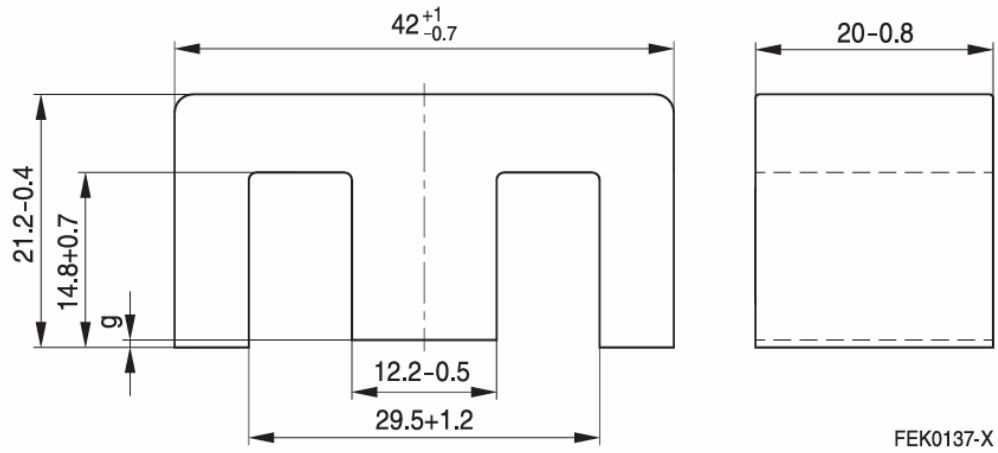


Figure 2.1: Dimensions considered for footprint

V_{in}	V_{out}	D	ΔV_{in}	C_{in}
48 V	64 V	0.250	0.96 V	14.27 μ F
	85 V	0.435	0.96 V	24.8426 μ F
62 V	64 V	0.031	1.24 V	1.06 μ F
	85 V	0.271	1.24 V	9.27 μ F

Table 2.6: Input Capacitance Values at Boundary Operating Points

The worst case value is selected and rounded up to the nearest standard capacitor value of $C_{in} = 25\mu\text{F}$.

Output Capacitor

The output capacitor is sized to limit the peak-to-peak output voltage ripple to 2% of the output voltage. From the charge balance on the capacitor:

$$C \geq \frac{I_{out} \cdot D}{\Delta V_{out} \cdot f_{sw}} \quad (2.8)$$

where $\Delta V_{out} = 0.02 \times V_{out}$ and $I_{out} = P_{out}/V_{out}$. The capacitance is evaluated at all boundary conditions and the results are listed in Table 2.7.

V_{in}	V_{out}	D	I_{out}	C
48 V	64 V	0.250	3.125 A	7.63 μF
	85 V	0.435	2.353 A	7.53 μF
62 V	64 V	0.031	3.125 A	0.95 μF
	85 V	0.271	2.353 A	4.67 μF

Table 2.7: Output Capacitance Values at Boundary Operating Points

The worst case capacitance of $C = 7.63 \mu\text{F}$ is selected, and a standard value of $C = 10 \mu\text{F}$ is chosen for the design.

2.2.5 MOSFET Ratings

The MOSFETs carries current during the ON interval and OFF interval. Considering only the lower side MOSFET, its RMS current is given by:

$$I_{SW,rms} = I_{in} \sqrt{D} \quad (2.9)$$

The blocking voltage of the MOSFET equals the output voltage V_{out} . Applying a 30% voltage derating factor:

$$V_{DS} \geq V_{out,max} \times 1.3 = 85 \times 1.3 = 110.5 \text{ V} \implies \boxed{\text{Use 150V rated MOSFET}} \quad (2.10)$$

The MOSFET ratings at boundary conditions are listed in Table 2.8.

The worst case RMS current is $I_{SW,rms} = 2.894 \text{ A}$.

To ensure symmetry and easy fabrication both the MOSFETs are chosen with the same ratings, and with an increased current rating.

$$\implies \boxed{\text{Use 10A rated MOSFET}}$$

Based on these requirements, a TO-247 three-legged MOSFET **SUG80050E** is selected. The device provides a voltage rating of 150V and a current rating significantly higher than the design requirement, ensuring safe operation.

V_{in}	V_{out}	D	$I_{SW,rms}$
48 V	64 V	0.250	2.193 A
	85 V	0.435	2.894 A
62 V	64 V	0.031	0.600 A
	85 V	0.271	1.766 A

Table 2.8: MOSFET RMS Current at Boundary Operating Points

2.2.6 Snubber Design

The snubber capacitor is selected based on the MOSFET output capacitance using the guideline:

$$C_s \approx 5 \times C_{oss} \quad (2.11)$$

where C_{oss} is the MOSFET output capacitance. This ensures that the snubber capacitor dominates the parasitic capacitance and effectively reduces voltage ringing. Based on this, the selected value is: $C_s = 2.2\text{nF}$

The snubber resistor is chosen to provide adequate damping by matching the characteristic impedance of the parasitic LC network:

$$R_s \approx \sqrt{\frac{L_p}{C_s}} \quad (2.12)$$

where L_p is the equivalent parasitic inductance. The resistor is implemented using four $6.8, \Omega$ resistors (2512 package) in series. The dissipation across each resistor is limited to 0.5W

$$R_s = 4 \times 6.8 = 27.2, \Omega \quad (2.13)$$

The power dissipated in the snubber is estimated as:

$$P_s = C_s V^2 f_{sw} \quad (2.14)$$

Substituting worst-case values ($V = 85, \text{V}$, $f_{sw} = 80, \text{kHz}$):

$$P_s = 2.2 \times 10^{-9} \times (85)^2 \times 80 \times 10^3 \approx 1.27\text{W} \quad (2.15)$$

The total power is distributed across four resistors:

$$P_{res} = \frac{1.27}{4} \approx 0.32\text{W} \quad (2.16)$$

Since each resistor is rated at 0.5W, the design operates within safe limits.

2.2.7 Miscellaneous Calculations

Fuse Rating

The fuse is rated for $\times 1.25$ the input and output currents of the converter. The scaled input and output currents are given in Table. 2.9.

I	Fuse Rating
4.386A	5.482A
3.396A	4.245A

Table 2.9: Fuse Ratings

The input fuse rating is selected as **6A**, and the output fuse rating is selected as **4A**.

MOV Rating

The MOV is rated for $\times 2$ the maximum input and output voltages of the converter. The scaled input and output voltages are given in Table. 2.10.

V	MOV Rating
62V	124V
85A	170V

Table 2.10: MOV Ratings

The input MOV rating is selected as **180V**, and the output MOV rating is selected as **120V**.

Bleeder Resistor

The bleeder resistor is designed to ensure safe discharge of the output capacitor within a specified time constant of 10s. The values are calculated in Table. 2.11.

C	R_{bleed}
10 μ F	1M Ω
25 μ F	400k Ω

Table 2.11: Bleeder Resistor Values

The input capacitor R_{bleed} value is selected as **1M Ω** , and the output capacitor R_{bleed} value is selected as **680k Ω** from the 1206 package.

Minimum Load Resistor

The resistor value is selected based on the maximum allowable power dissipation. For a 2512 resistor, the power rating is limited to 0.5, W. Using the worst-case output voltage:

$$P_{bleed} = \frac{V_{out,max}^2}{R_{bleed}} \quad (2.17)$$

Rearranging:

$$R_{bleed} \geq \frac{V_{out,max}^2}{P_{max}} = \frac{85^2}{0.5} = 14450\Omega \quad (2.18)$$

A standard value: $\Rightarrow$ Use $R_{bleed} = 18k\Omega$

The power dissipated is:

$$P_{bleed} = \frac{85^2}{18000} \approx 0.401W \quad (2.19)$$

which is within the safe operating limit of the 2512 package.

Use of Low-ESR Ceramic Capacitors

In addition to the bulk input and output capacitors, low-ESR ceramic capacitors of 0.1, μ F 1206 X7R 150V are placed in parallel across both the input and output. These capacitors provide a low impedance path for high-frequency switching currents and help suppress voltage spikes and noise that cannot be effectively filtered by bulk capacitors due to their higher ESR.

By placing the ceramic capacitors close to the switching nodes, high-frequency ripple and electromagnetic interference (EMI) are reduced.

2.3 Sensing Circuit Calculations

2.3.1 Voltage Sensor

A voltage divider is used across the input and output terminals to measure the voltage. The 2% voltage ripple and a 5% margin is multiplied with the maximum values. The required voltages to measure is given in Table. 2.12.

The output of the sensor is connected to an AMC1300 isolation op-amp which requires an input of $\pm 0.25V$.

The resistor package used is 0805, with the voltage and power ratings limited to 60% and 50% of their rated values respectively. This is equal to $V_{package} = 60V$ and $P_{package} = 0.05W$. The resistor value considered is:

$$R_{package} = \frac{V^2}{P} = \frac{60^2}{0.05} = 72k\Omega \quad (2.20)$$

V_{max}	V_t
62V	67V
85V	92V

Table 2.12: Voltage to Measure Values

The standard value of $R_{\text{package}} = 68k\Omega$ is chosen.

Input Voltage

The output is 0.25V, the voltage across the 0805 heavy resistance network will be 66.75V. The total resistors in series required:

$$n = \frac{66.75}{60} = 1.11 \approx \mathbf{2nos} \quad (2.21)$$

The value of R_m is calculated as follows:

$$\frac{R_m}{R_m + 2 \times 68k} \times 67 = 0.25 \quad (2.22)$$

$$\Rightarrow R_m = 509\Omega \approx \mathbf{470\Omega} \quad \text{Standard Value} \quad (2.23)$$

2.3.2 Current Sensor

A shunt divider network is used where the output voltage of the network is proportional to the current flowing through it. The 20% current ripple and a 5% margin is multiplied with the maximum value to obtain 6A.

The resistor package used is 2512, with the power rating limited to $P_{\text{package}} = 0.5W$. The resistor value considered with m numbers in parallel is calculated from:

$$6 \times \frac{R_{\text{package}}}{m} = 0.25 \quad (2.24)$$

and

$$\frac{6^2}{m} \times R_{\text{package}} = 0.25 \quad (2.25)$$

On solving the above two equations we obtain, $R_{\text{package}} = 100m\Omega$ and $m = 3$.

2.3.3 R-C Filter

We are required to obtain a -40dB/dec , with a bandwidth of $\frac{f_{sw}}{2}$. The first R-C filter is placed at the input of the isolation op-amp, and the second is obtained using a differential op-amp.

Voltage Sensors

$$R_{th_{Vin}} = 470 || (2 \times 68k) = 468.3813\Omega \quad (2.26)$$

$$C_{Vin} = \frac{1}{2\pi \times 40k \times 468.38} = 8.49nF \approx \mathbf{8.2nF} \quad \text{Std Value} \quad (2.27)$$

$$R_{th_{Vout}} = 470 || (3 \times 68k) = 468.9196\Omega \quad (2.28)$$

$$C_{Vout} = \frac{1}{2\pi \times 40k \times 468.919} = 8.485nF \approx \mathbf{8.2nF} \quad \text{Std Value} \quad (2.29)$$

Current Sensor

Two identical resistors are connected in series with the output of the shunt sensor to reduce the size of the filter capacitor.

$$R_{th_{In}} = \frac{100m}{3} + 20 = 20.033\Omega \quad (2.30)$$

$$C_{In} = \frac{1}{2\pi \times 40k \times 20.033m} = \mathbf{220nF} \quad \text{Std Value} \quad (2.31)$$

2.3.4 Op-Amp Calculations

A quad Op-Amp IC LM324 was used to implement the signal conditioning circuit. The network is identical for the all the sensors due to the presence of AMC1300 isolation op-amp.

The input to the op-amp is **2.05V** with the output being **3.3V**.

Selecting **R₁ = 10kΩ**, **R₂** can be calculated as:

$$\frac{R_2}{R_1} = \frac{3.3}{2.05} = 1.609 \quad (2.32)$$

Final value obtained will be, **R₂ = 15kΩ**.

C₂ is calculated as:

$$C_2 = \frac{1}{2\pi \times 40k \times 15k} = 258.78pF \approx \mathbf{270nF} \quad \text{Std Value} \quad (2.33)$$

2.3.5 ADC Input Filter

The impedance matching resistor required by the ADC is **R_{adc} = 56Ω**, considering a bandwidth of **500kHz**, **C_{adc}** can be calculated as:

$$C_{adc} = \frac{1}{2\pi \times 500k \times 56} = \mathbf{5.6nF} \quad \text{Std Value} \quad (2.34)$$

2.4 PCB Considerations

Table. 2.13 gives the track widths considered for the different sections.

A minimum clearance of 0.3mm was considered based on the maximum voltage of 92V across the circuit, with a breakdown voltage of 30kV/cm.

Region	Width (mils)
Power	118
Snubber	50
Gate Driver	20
Current Shunt Output	15
Control	20

Table 2.13: Track Width Values

2.5 Controller Calculations

The controller is designed at the worst case operating point, which corresponds to the minimum input voltage and maximum output voltage. The input capacitance is not considered in the design, only the output capacitance.

Parameter	Value
V_{in}	48 V
V_{out}	85 V
D	0.4353
$1 - D$	0.5647
R	36.125 Ω
L	0.309 mH
C	10 μF
r_L	10 m Ω

Table 2.14: Worst Case Operating Point for Controller Design

2.5.1 DC Operating Point

The DC inductor current and capacitor voltage are obtained from Eq. 1.22:

$$I_L = \frac{V_{in}}{r_L + (1 - D)^2 R} = \frac{48}{0.01 + (0.5647)^2 \times 36.125} = \frac{48}{11.533} = \boxed{4.159 \text{ A}} \quad (2.35)$$

$$V_C = \frac{(1-D)R \cdot V_{in}}{r_L + (1-D)^2 R} = \frac{0.5647 \times 36.125 \times 48}{11.533} = \boxed{85 \text{ V}} \quad (2.36)$$

2.5.2 Control-to-Output Transfer Function

Substituting the operating point values into Eq. 1.32, the numerator and denominator coefficients are computed as follows.

Numerator Coefficients

The constant term in the numerator is:

$$(1-D)V_C - I_L r_L = (0.5647 \times 85) - (4.162 \times 0.01) = \boxed{47.983} \quad (2.37)$$

The coefficient of s in the numerator is:

$$-LI_L = -0.309 \times 10^{-3} \times 4.162 = \boxed{-1.285 \times 10^{-3}} \quad (2.38)$$

Denominator Coefficients

The coefficient of s^2 in the denominator is:

$$LC = \boxed{3.09 \times 10^{-9}} \quad (2.39)$$

The coefficient of s in the denominator is:

$$\frac{L}{R} + Cr_L = \frac{0.309 \times 10^{-3}}{36.125} + 10 \times 10^{-6} \times 0.01 = \boxed{8.654 \times 10^{-6}} \quad (2.40)$$

The constant term in the denominator is:

$$\frac{r_L + (1-D)^2 R}{R} = \frac{0.01 + 11.523}{36.125} = \frac{11.533}{1.116 \times 10^{-7}} = \boxed{0.32} \quad (2.41)$$

Resulting Transfer Function

The control-to-output transfer function at the worst case operating point is therefore:

$$G_{vd}(s) = \frac{47.983 - 1.285 \times 10^{-3}s}{3.09 \times 10^{-9}s^2 + 8.654 \times 10^{-6}s + 0.32} \quad (2.42)$$

2.5.3 Loop Shaping Design

Loop Shaping Procedure

The loop shaping method designs the PI controller by evaluating the plant at the desired gain crossover frequency ω_c and computing the required compensator gain and phase. The procedure is as follows:

1. Evaluate the plant transfer function at $s = j\omega_{gc}$:

$$G_{vd}(s) = \frac{47.983 - 1.285 \times 10^{-3}s}{3.09 \times 10^{-9}s^2 + 8.654 \times 10^{-6}s + 0.32} \quad (2.43)$$

2. Compute the plant magnitude and phase:

$$K_C = \frac{1}{|G_{vd}(j\omega_{gc})|}, \quad \angle G_{vd}(j\omega_{gc}) = \arctan\left(\frac{\text{Im}(G_{vd})}{\text{Re}(G_{vd})}\right) \quad (2.44)$$

3. Compute the required compensator phase:

$$\delta_c = PM - 180^\circ - \angle G_{vd}(j\omega_{gc}) \quad (2.45)$$

4. Extract the PI controller parameters:

$$K_p = K_c \cos(\delta_c), \quad K_i = -K_c \cdot \omega_c \cdot \sin(\delta_c) \quad (2.46)$$

MATLAB Function Implementation

The following MATLAB script implements the loop shaping procedure, sweeping over phase margin values from 40 to 80 and crossover frequencies from $\omega_{sw}/500$ to $\omega_{sw}/50$, and returning only physically valid solutions where $K_p > 0$, $K_i > 0$:

Listing 2.1: Loop Shaping Controller Tuning Script

```
function cntrlTuning()
clc
for i = 40:5:80
    for j = 50:5:500
        Wgc = 2*pi*80e3/j;
        X = 1j * Wgc;

        % Plant transfer function G_vd(s)
        num = -1.285e-3*X + 47.983;
        denom = 3.09e-9*X*X + 8.654e-6*X + 0.32;
        G1 = num/denom;

        magnG1 = abs(G1);
```

```

argG1 = angle(G1) * 180/pi;

KC    = 1/magnG1;
delta = i - 180 - argG1;

KP = KC * cosd(delta);
KI = - KC * Wgc * sind(delta);

% Display only physically valid solutions
if (KP > 0) && (KI > 0)
    STR = "PM:" + i + " Wgc: " + round(Wgc) + ...
          " j: " + j + " Kp: " + KP + " Ki: " + KI;
    disp(STR);
end
end
disp(newline)
end
end

```

Selected Controller Values

The script is run and the output is inspected for solutions that satisfy all constraints. The selected operating point corresponds to $PM = 45$ and $j = 50$, giving a crossover frequency of:

$$\omega_c = \frac{2\pi \times 80000}{50} = 10053 \text{ rad/s} \quad (f_c = 1600 \text{ Hz}) \quad (2.47)$$

Parameter	Expression	Value
Proportional Gain	K_p	0.0014399
Integral Gain	K_i	10.134
Gain Crossover Frequency	ω_c	10053 rad/s (1600 Hz)
Phase Margin	PM	45

Table 2.15: Selected PI Controller Parameters

The final PI controller transfer function is:

$$C(s) = K_p + \frac{K_i}{s} = 0.0014399 + \frac{10.134}{s} \quad (2.48)$$

3. Results

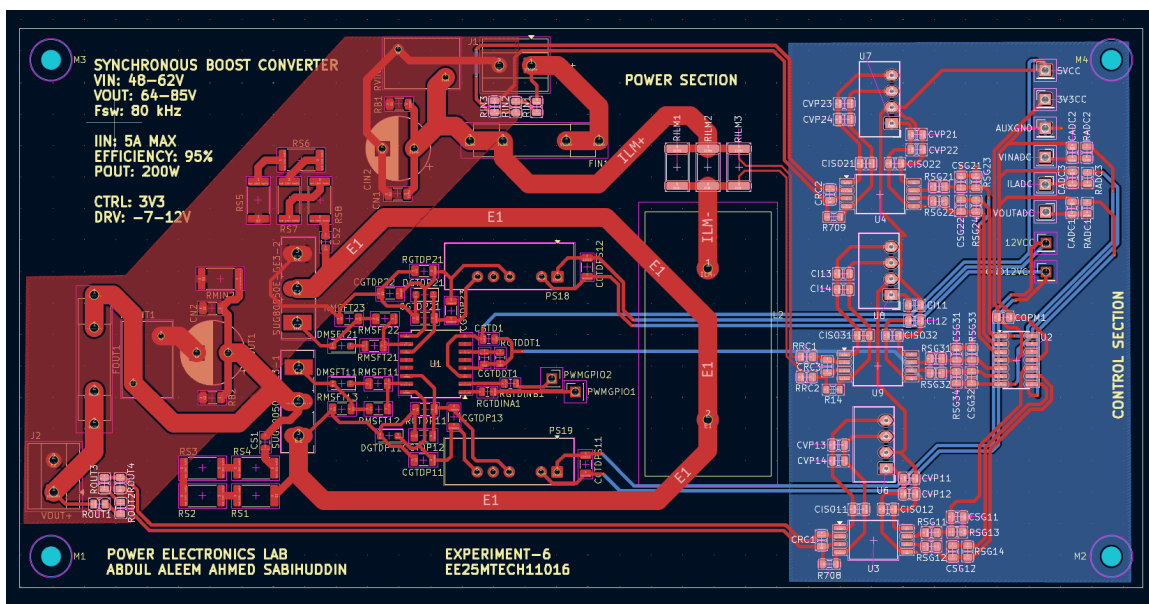


Figure 3.1: Final PCB Layout

4. Learnings/Observations

1. PCB layout plays a critical role in converter performance. Minimizing high-current loop areas and proper placement of components such as input capacitors and MOSFETs is essential to reduce switching noise and voltage spikes.
2. The switching node is a major source of high-frequency noise, and its area must be minimized to limit electromagnetic interference (EMI).
3. The use of low-ESR ceramic capacitors in parallel with bulk capacitors significantly improves high-frequency filtering and reduces voltage ripple.
4. Bleeder resistors are required to ensure a controlled discharge of the output capacitor under no-load conditions. Conservative design choices, such as over-rating components and using simplified design equations, improve robustness and ensure safe operation under varying conditions.
5. The placement of the gate driver relative to the MOSFET significantly affects switching performance. Keeping the gate loop short reduces parasitic inductance, prevents oscillations, and improves switching transitions.
6. Isolation in sensing circuits significantly enhances noise rejection, especially in high dv/dt environments such as switching converters, it also protects the control circuitry from any faults at the power converter side.
7. The use of differential amplifiers in the sensing circuit enables accurate measurement of voltage signals in the presence of high common-mode noise. Moreover, it has minimal loading effect at the ADC input terminals as well.